CS190C Lec2

From Seq2Seq to Transformer

Overview

- What is Seq2Seq
- Seq2Seq with RNN
- Seq2Seq with Transformer
  - Attention
  - Position Embedding
  - Feed-Forward Network
  - Residual Connection
  - The Whole Transformer

PART1: What is Seq2Seq

Examples

- Given a Chinese sentence, translate to English $\Rightarrow$ Machine translation
- Given a long paragraph, conclude to a short sentence $\Rightarrow$ Abstract writing
- Given an academic report, convert to easy-to-understand article $\Rightarrow$ Style transfer

Conclusion

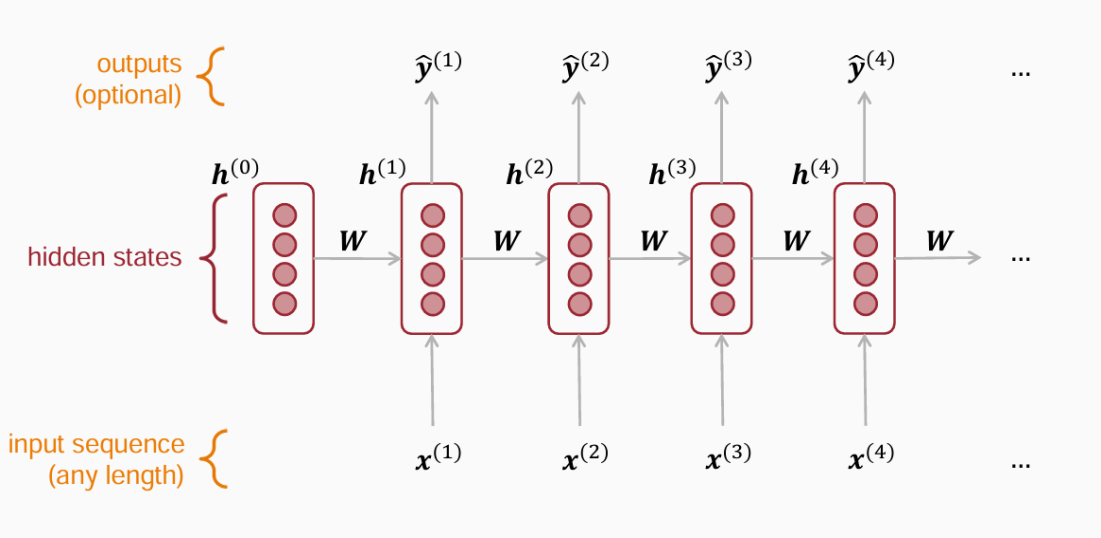
Seq2seq (aka. Sequence to sequence) is a kind of language task to input a sequence and output a sequence according to certain demand.

PART2: Seq2Seq with RNN

RNN Review

`

- At each time step, the model receives and encodes a new word.
- Combines this newly encoded word with the context of the previous text.
- With each time step, the representation (embedding) of the text is continually updated.



`

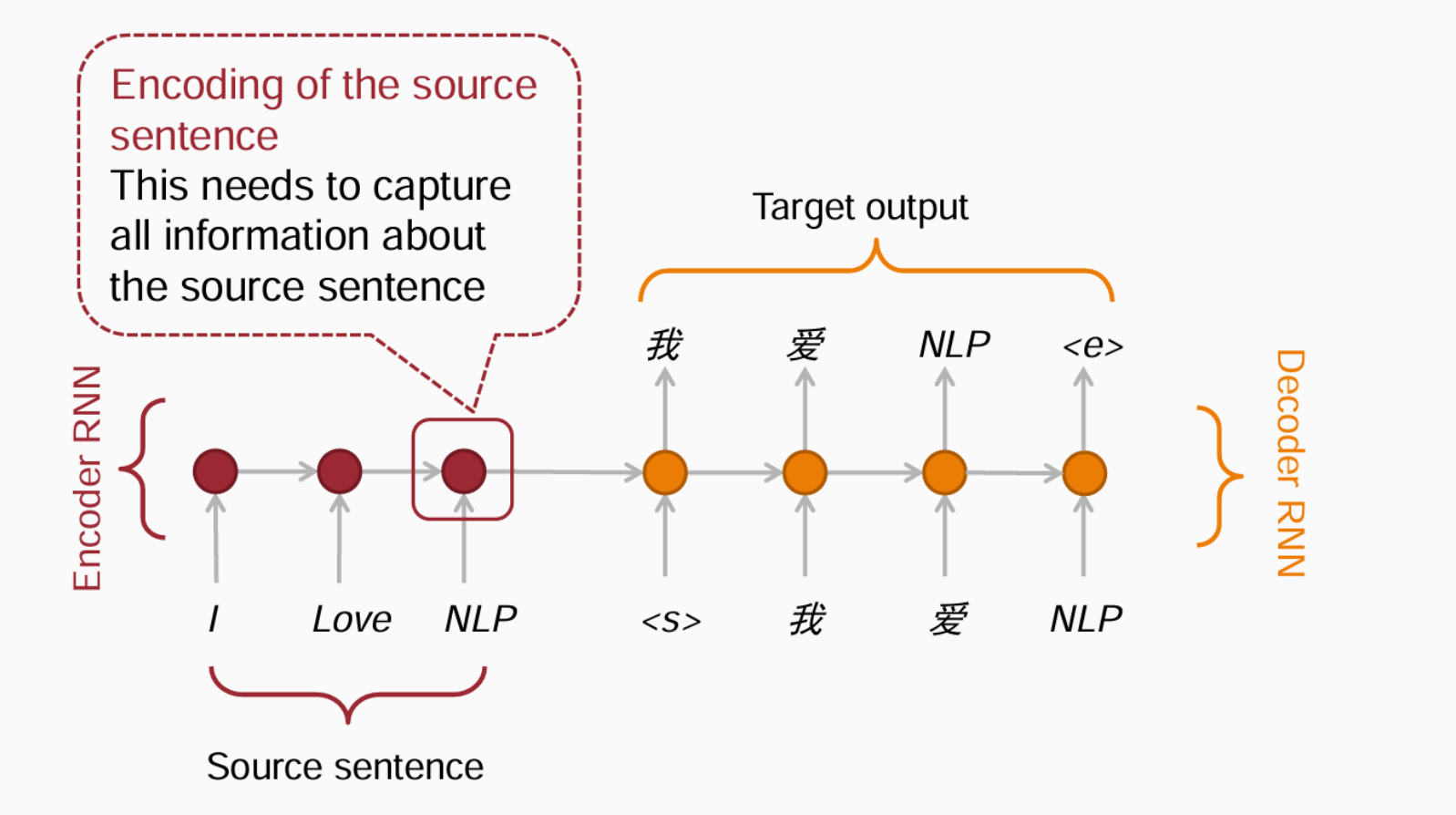
- At the final time step, this embedding of text contains the complete information of the entire sequence.

A Basic Approach

- The embedding of text contains the complete information of the entire sequence at last...
- Input a sequence and use RNN (**Encoder**) to process it...
- The final embedding h^T is expected to capture all the information from the text.
- We can further generate the output based only on h^T.
- Use another RNN (**Decoder**) for generation, immediately following the first RNN.

Seq2Seq with RNN

- **Encoder and Decoder**



Pros and Cons?

- Pros: Simple and intuitive.
- Gradient problems from RNN.
- For long sequences, the model may not perform well —— not only gradient problems...
  - Encode everything into a single vector without proper weights.
  - In forward: $h^t = \sigma(W_h h^{t-1} + W_e e^t + b_1)$
  - Earlier information is repeatedly overwritten and diluted, even if it is important.
  - e.g. `The writer of the books _____(is/are)`
    - $\Rightarrow$ `books` is closer and `writer` is farther.

PART3: Seq2Seq with Transformer

Attention Is All You Need

PART3.1: Attention

How to encode with proper weight?

```
The writer of the books _____(is/are)
```

To fill the blank, we should find words whose properties strongly correlate with its "problems".

- **"Problems"**: How do we describe what a word is looking for?
- **"Properties"**: How do we describe the attributes of the word itself?
- **"Attention"**: If a word's "problem" and another word's "property" strongly match, it will cause significantly high weight.
- Once a match is found, how do we represent the actual information the target word provide?

How to encode with proper weight?

- All above should rely on the word embeddings.
  - If a word has embedding $x \in \mathbb{R}^d$, each x_i ($1 \leq i \leq d$) indicates the magnitude of a specific semantic feature.
- The whole semantic of a word can be seen as conbination of all semantic features and their significance level.
- Each semantic feature may represent certain kind of problems, properties and information.

How to encode with proper weight?

For all d semantic features, we train three distinct weight matrices:

- Matrix $W^Q \in \mathbb{R}^{d \times d_k}$
  - Certain semantic feature causes a "problem", whose embedding $\in \mathbb{R}^{d_k}$.
- Matrix $W^K \in \mathbb{R}^{d \times d_k}$
  - Certain semantic feature will represent a "property", whose embedding $\in \mathbb{R}^{d_k}$.
- Matrix $W^V \in \mathbb{R}^{d \times d_v}$:
  - Certain semantic feature will feed back certain kind of information, whose embedding $\in \mathbb{R}^{d_v}$.

How to encode with proper weight?

- How do we calculate the "problem" of a word?
- Which is the combination of semantic features?
- Recall: Certain value of x_i means the significance level of i-th semantic features.

$$W^Q = [w_1^q, w_2^q, \dots w_d^q]^T$$
$$q = x_1 w_1^q + x_2 w_2^q + \cdots + x_d w_d^q$$

- $\implies q = xW^Q$, which is the "problem" of a word.
- Similarly: $k = xW^K, v = xW^V$
- Embedding x is projected into q, k, v.

Scaling Dot-Product Attention (SDPA)

- For word i with "problem" q_i, word j's weight score with word i is $q_i k_j^T$.
- $\mathrm{Softmax}$ function: Normalize the scores across all words and creates a probability distribution, as the weight for weighted sum v_j fed back to word i.
- In matrix formula:

$$\boxed{\mathrm{Attention}_i = \mathrm{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V}$$

- Tips: $\sqrt{d_k}$ is a scaling factor:
  - As dimensions grow, dot products can get extremely large.
  - Ensure the distribution not degenerate to one-hot.
  - Pushing Softmax into regions with near-zero gradients in backward.

Multihead Attention (MHA)

- Does a semantic feature have only one "problem", only one "property"?
- Repeat SDPA in H channels, each channel stands for certain kind of "problem" and "property" pair.
- So word i has: h_1,h_2,...,h_H ($h_j = \mathrm{Attention}_i^j$).

$$\mathrm{Attention}_i = \mathrm{Concat}(h_1, h_2, \ldots, h_H)W^O$$

PART3.2： Position Embeddings

A Problem

- If two sentences have same word but in different orders, they usually has different meanings.
  - `I study LLM`, `LLM study I`
- But what about MHA?
  - Only word embeddings and their q, k, v matters, do not include position.

$\Longrightarrow$ We should add some "tag" for each word, representing their position in the sentence. Thus it will have an impact on MHA.

Sinusoidal position encoding in Transformer 2017

We should bring a regular transformation to word embeddings x, but how?

- Actually the transformation should be a function of position.

But is that enough?

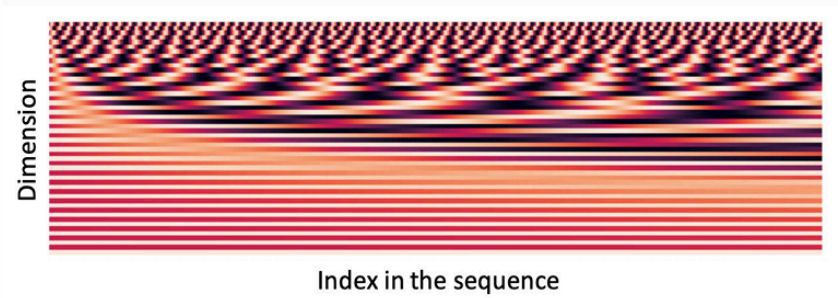
- Certain semantic features might be sensitive to local word order, but some are not.
  - The transformation must also be a function of the embedding dimension.
- The mapping function must output a unique vector for every exact (position, dimension) pair to prevent any ambiguity.

Sinusoidal position encoding in Transformer 2017

Let p_i be the position embedding at position i with dimension d.

$$p_j = \begin{cases} \sin(pos \cdot \omega_j), & \text{if } j \text{ is even} \\ \cos(pos \cdot \omega_{j-1}), & \text{if } j \text{ is odd} \end{cases}, \text{ where } \omega_d = \frac{1}{10000^{j/d}}$$

- $\text{Period} = \frac{2\pi}{\text{pos}} 10000^{j/d}$
- For different position, there will be a different embedding function $\Rightarrow$ one-to-one mapped.
- For different dimension: the period will change significantly from 2π to 20000π, standing for different sensitivity of different kind of semantics.



Pros and cons?

- It may also represent some relative position difference in a way.

$$\begin{bmatrix}\sin((pos+k)\omega_j)\\ \cos((pos+k)\omega_j)\end{bmatrix} = \begin{bmatrix}\cos(\omega_j k) & \sin(\omega_j k)\\ -\sin(\omega_j k) & \cos(\omega_j k)\end{bmatrix}\begin{bmatrix}\sin(pos\cdot\omega_j)\\ \cos(pos\cdot\omega_j)\end{bmatrix}$$
$$= M_{k,j}\begin{bmatrix}\sin(pos\cdot\omega_j)\\ \cos(pos\cdot\omega_j)\end{bmatrix}$$
$$\implies p_{pos+k} = M_k \cdot p_{pos}, \text{ where } M_k = \mathrm{diag}(M_{k,1}, M_{k,2}, \ldots, M_{k,d/2})$$

- Actually, instead of absolute position, in natural language, difference in semantic caused by position is usually relative position.

Pros and cons?

- But sinusoidal position encoding is still absolute position.

For calculation of q_i and k_j:

$$\begin{aligned} q_i k_j^T &= (x_i + p_i) W_Q W_K^T (x_j + p_j)^T \\ &= x_i W_Q W_K^T x_j^T + p_i W_Q W_K^T x_j^T + x_i W_Q W_K^T p_j^T + p_i W_Q W_K^T p_j^T \end{aligned}$$

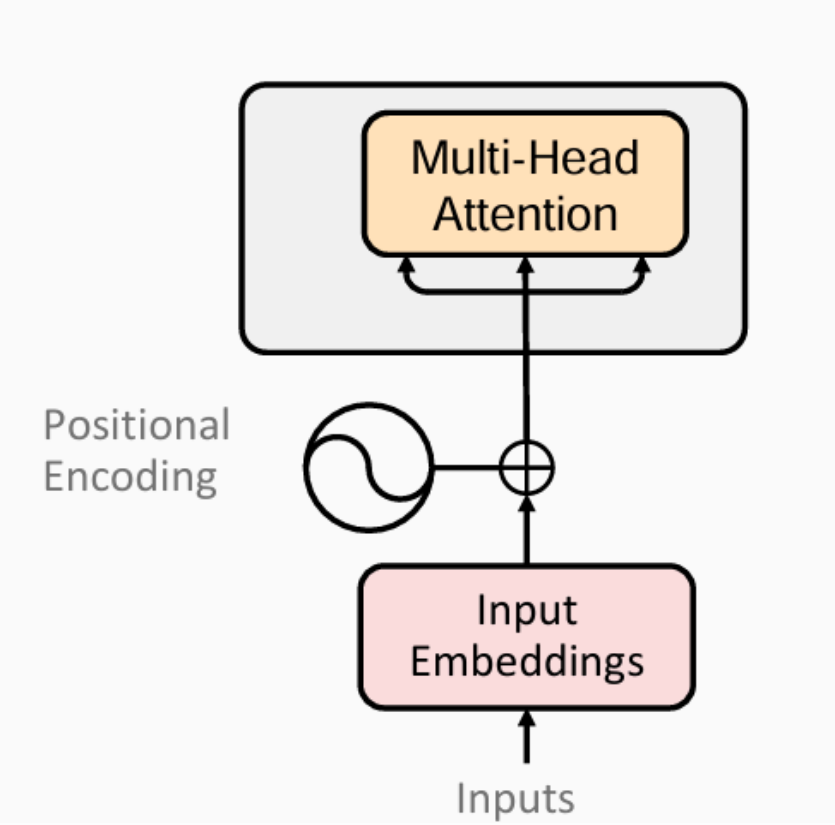
- For $p_i W_Q W_K^T p_j^T$, if $W_Q W_K^T$ does not satisfy the structure of M_k, it is related to absolute position.
- For $p_i W_Q W_K^T x_j^T$ and $x_i W_Q W_K^T p_j^T$, it is completely determined by absolute position.

What is the disadvantage of absolute position?

- Difference in semantics usually doesn't depend on it.
  - But it is different for attention calculation given a pair of tokens with the **same relative position** and **different absolute position**.
  - The model may unnecessarily memorize absolute positions.
- When the model encounters very long sentences in reasoning after training,
  - The model doesn't know how to process the new, unseen absolute positions because it wasn't trained on them.
  - The outcomes will worsen.

So most LLMs now use `RoPE` to embbed position, which is a relative position embedding method (In Lec3).

The language model for encoding so far...



- Add position embeddings before enter the encoder block.
- Enter the encoder block, using Multi-Head Attention to enrich the word embeddings using the information of text.
- We can use multiple layers of blocks to enrich it multiple times.

- But there still exists some problems...

Problems

- MHA consists mostly of linear combinations. It struggles to process complex, non-linear feature transformations.
- Moreover, output embeddings can only aggregate information in the context.

Example: `I twisted the door handle, and the door _____`

To answer `opened`, the model should learn:

- Common sense of physics: Twisting will make it "open" instead of "broken".
- Common sense of life: We use handle to open it instead of threshold.
- Other "common sences" outside the text and beyond MHA.

How can we fix it?

PART3.3: Feed-Forward Network

Up-Projection

Suppose there are a numerous latent "global features".

- e.g. "Is it a tool to open something?"
- The number of features is larger than embedding dimensions.

How can we measure the relationship between the word embedding and this feature?

- Dot product: xk_i^T is the relationship of x and feature i.
- Activation function σ: Introduce non-linear.
  - e.g. ReLU: $\mathrm{ReLU}(x) = \max(x, 0)$
- **Up-Projection**: $x_{ff} = \sigma(xW_1)$
  - Represents "the relationship between word embedding and different features."

Down-Projection

We must project the vector back to the model's original dimension.

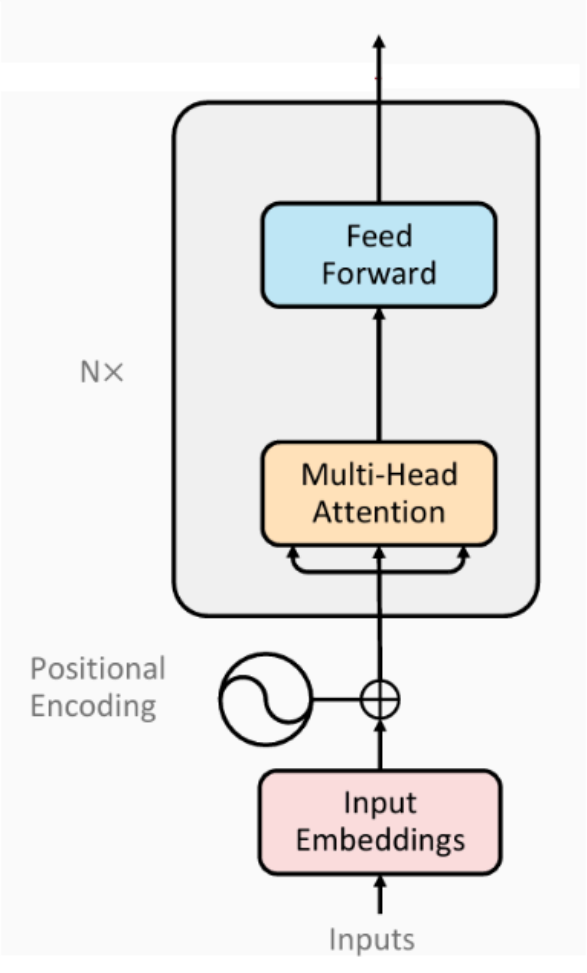
- **Down-Projection**: $x_{output} = x_{ff} W_2$.
  - Maps the high-dimensional patterns back into the embedding space.
- Full FFN:

$$\boxed{x_{output} = \sigma(xW_1) \cdot W_2}$$

- Dimension of x_{ff} is usually $4d$.

Further Reading: Transformer Feed-Forward Layers Are Key-Value Memories

The language model for encoding so far...



- **Self-Attention**: Captures contextual information from the text.
- **Feed-Forward Networks (FFN)**: Extracts global feature information.
- Repeat the above steps for N layers to deepen representation.
- However, as N increases, training becomes more difficult.
  - Hard to train beyond 20~30 layers for such a kind of model.

How can we address this challenge?

PART3.4: Residual Connection

The Forward Pass Perspective: A large number of layers?

- Consider a very deep network with N layers, we actually need several layers to be expressive enough.
- So it means: If N is very large, some layers can act as an identity transformation to maintain the information.
- However, it is difficult for MHA and FFN to learn a perfect Identity matrix I.
- So large N may leads to performance degradation on the contrary.

The Forward Pass Perspective: A large number of layers?

Question: It is hard to train I for a layer. Is it hard to train 0 ?

$\Rightarrow$ With $l_2 - \mathrm{Norm}$ (weight decay), parameters naturally tend toward 0.

- So why don't we learn the parameter based on identity transformation I?
  - Only learn difference of information, not absolute information.

$$y_l = \mathrm{Attn}_l(x_l) \;\Rightarrow\; y_l = \mathrm{Attn}_l(x_l) + x_l$$
$$x_{l+1} = \mathrm{FFN}_l(y_l) \;\Rightarrow\; x_{l+1} = \mathrm{FFN}_l(y_l) + y_l$$

It can significantly avoid the degradation caused by large N.

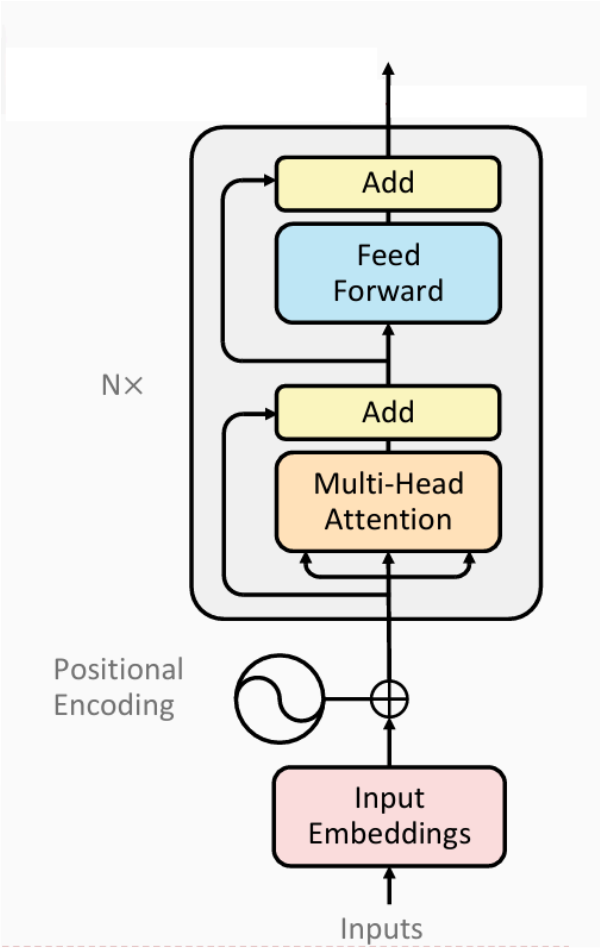
The Backward Pass Perspective: Chain rule?

$$\frac{\partial L}{\partial x_l} = \frac{\partial L}{\partial x_{l+1}} \cdot \frac{\partial x_{l+1}}{\partial y_l} \cdot \frac{\partial y_l}{\partial x_l}$$
$$= \frac{\partial L}{\partial x_{l+1}} \cdot \mathrm{Attn}_l' \cdot \mathrm{FFN}_l'$$

For large N, we multiply many terms to update parameters of bottom layers.

- Familiar? It is the same as RNN.
- Vanishing gradient may happen when some terms are small, which leads to almost not learnable parameters of bottom layers.

The Backward Pass Perspective: Chain rule?



Adding residual connection:

$$\frac{\partial L}{\partial x_l} = \frac{\partial L}{\partial x_{l+1}} \frac{\partial x_{l+1}}{\partial y_l} \frac{\partial y_l}{\partial x_l}$$
$$= \frac{\partial L}{\partial x_{l+1}} (I + \mathrm{Attn}_l')(I + \mathrm{FFN}_l')$$

- Even if some terms has gradient approches to 0, the gradients to flow directly from the top to the bottom.

PART3.5: Layer Normalization

Estimate the scale of activations

- For a certain activation tensor, we consider the values of its elements as a distribution.
- The mean of distributions is usually close to 0, but the variances are different. Therefore, we use variance to measure the overall magnitude of the activations.
- In a residual network, how does the variance of these activations change across different layers?

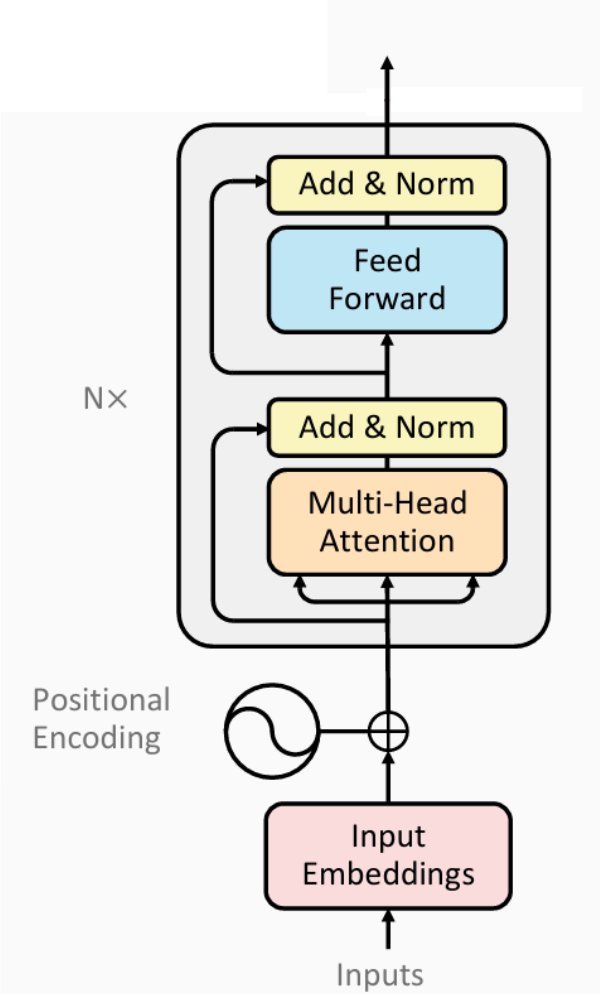
Estimate the scale of activations

- In preliminary stage of training, parameters are initialized randomly.
  - Activation of each module F is approximately independent of its input.

$$\mathrm{Var}(x_2) = \mathrm{Var}(x_1 + F(x_1)) = \mathrm{Var}(x_1) + \mathrm{Var}(F(x_1)) = (1+\alpha)\mathrm{Var}(x_1)$$
$$\mathrm{Var}(x_3) = \mathrm{Var}(x_2 + F(x_2)) = (1+\alpha)\mathrm{Var}(x_2) = (1+\alpha)^2\mathrm{Var}(x_1)$$
$$\mathrm{Var}(x_n) = (1+\alpha)^{n-1}\mathrm{Var}(x_1)$$

- Exponential Explosion!

Layer norm



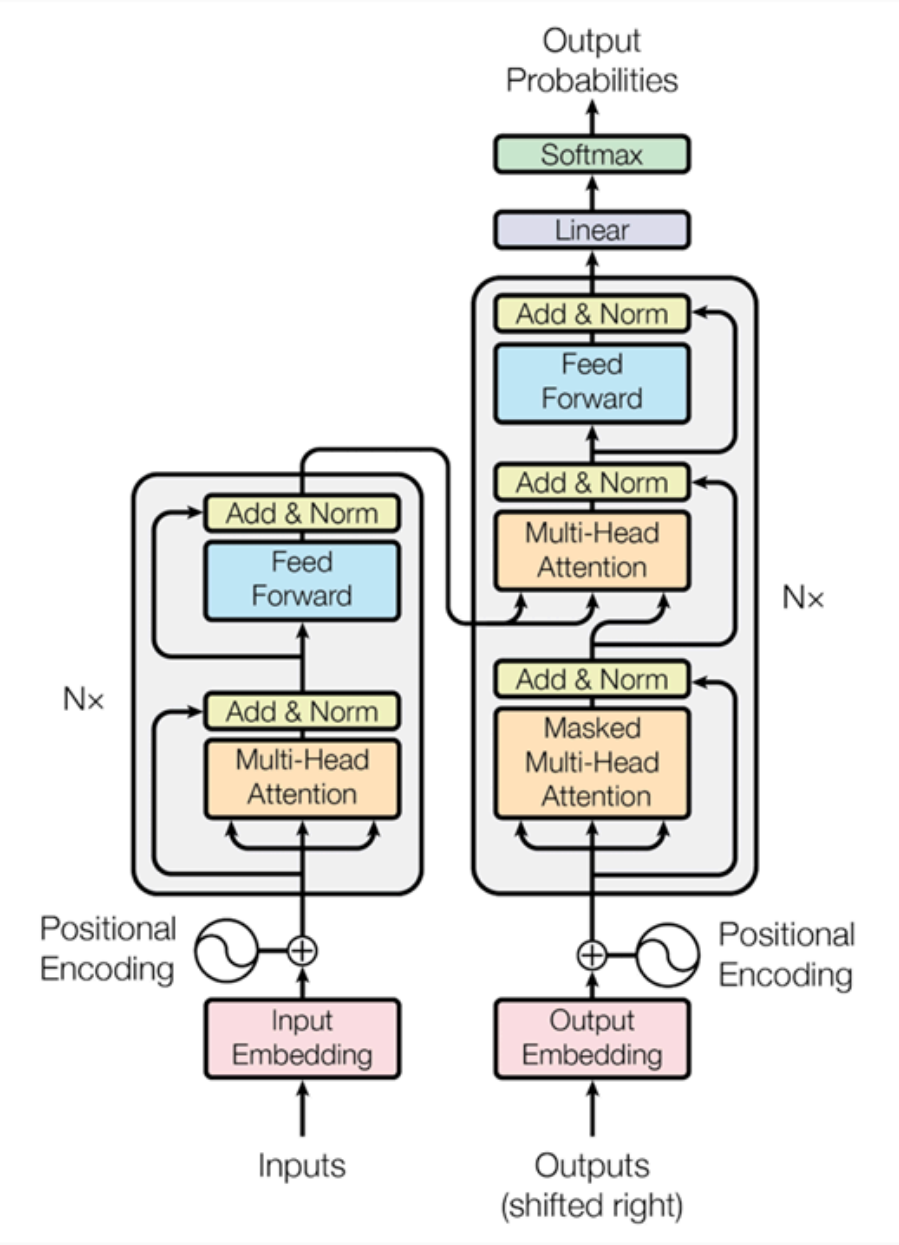
We should perform a normalization to activations in each module. That is:

$$\hat{x} = \frac{x - \mu}{\sigma}$$

It can reduce the explosion of activations.

In next lecture, we will discuss where to perform (post-norm & pre-norm), how to perform(LN & RMSNorm)

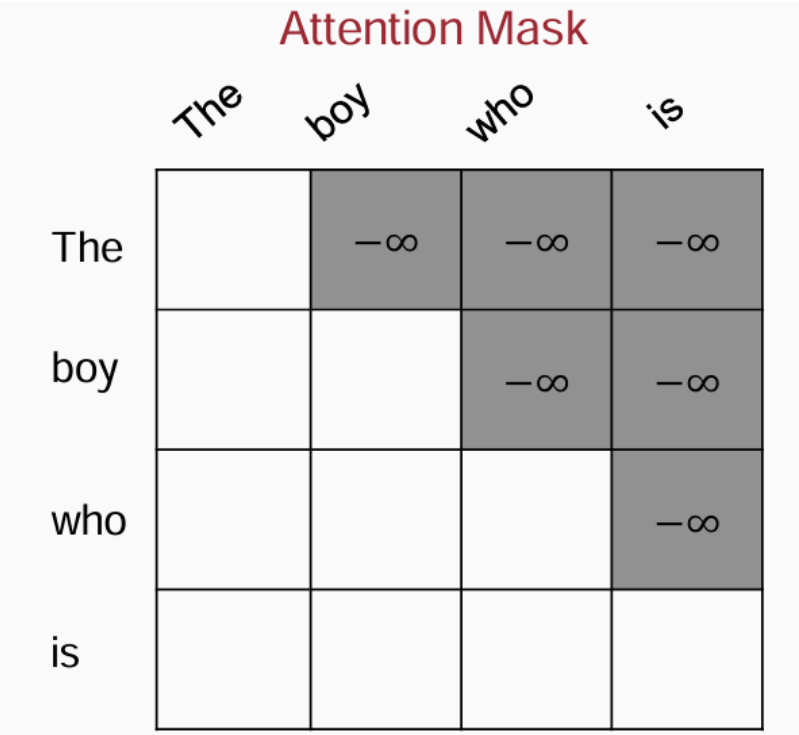
PART3.6: The Whole Transformer



Seq2Seq with Transformer

- Encoder
  - Use N layers to encode the input embeddings, the output of N-th layer is the encode outcomes.
- Decoder
  - Self-attention modules: absorb the decoded information.
  - Cross-attention modules: absorb the information of encoder.
- Linear
  - Projects decoder from embedding to distributions over vocabulary words.

What is Masked MHA?



$$a_{ij} = \begin{cases} q_i^T k_j, j \leq i \\ -\infty, j > i \end{cases}$$

For QK^T matrix:

- We apply an upper triangular mask to it, setting the upper-right elements equals to $-\infty$.
- When we apply $\mathrm{Softmax}$ to it, the upper part elements receive weight 0.
- It means: When i-th word aggregates information in the context, j-th word should contribute no information if $j > i$.

Why Masked MHA?

- What does the training data look like?
  - $\Rightarrow$ `inputs` + `outputs`
- Masking enforces the **auto-regressive property** during training.
  - During training process, the model decoder tries to output a sequence of embeddings, that is the decoded "sentence".
  - At the i-th position, the model uses the encoder output and the previously generated words (positions from 1 to i) to predict the $(i+1)$-th word.
  - Therefore, the mask can effectively prevents **information leakage**, forcing the model to actually learn next-token prediction.